1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L34 – Experiment Design - II**

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4 **Empirical Study Stages**

- Broadly, four stages
  - Identification of research question(s)
  - Determination of variables
  - Design of experiment
  - Analysis of empirical data

5 **Understanding the Stages**

- We discussed the first two stages
- In this lecture, we shall continue our discussion on remaining stages

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Experiment Design

7 **Why Important**

- Consider the example again
  - We identified a testable research question RQ3
  - We also have identified factors and levels – leading to two interfaces to study
  - What should we do now?

8 **Why Important**

- The easiest way out is to rely on your close friends (involves least hassles)
- You ask five of them to participate
- To each, you show the two interfaces and ask them to type
- Typing speeds are recorded and you get your empirical data

9 **Why Important**

- It sounds logical and straightforward
- Unfortunately, there are many issues with the seemingly straightforward approach

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Participants

11 **Participants - Profile**

- Your friends probably belong to same age group and similar background (education,

socio-economic conditions and cultural) - a *homogeneous* group
- Older people may have different typing behavior than teenagers
- It is also likely that typing speed is different between a regular computer user and an infrequent user of computers

12 **Participants - Profile**
- What we are talking of is popularly called the *user profile*

13 **Participants - Profile**
- User profile may be factors - may like to use them as independent variables

14 **Participants - Profile**
- There may be large number of such factors – large number of test conditions
- To keep our experiment simple, we may like to keep them out of consideration
- At the same time, we should take into account their effect on observation

15 **Participants - Profile**
- We can do so by treating them as control variables
- We fix their value for our experiment
  - We may decide to have teenagers, frequent users of computers
  - Your friends may fit into this profile

16 **Participants - Profile**
- Downside - you cannot claim the conclusion is applicable to *any* user
- You are concluding for only those users whose profile matches with participants'

17 **Participants - Number**
- Other important issue - number of participants
  - Is five a good number?
  - Should you go for a bigger number (say ten)?
  - Is that sufficient?

18 **Participants - Number**
- There is no definite answer
- According to one estimate, 5 participants good enough for an empirical research
- However, there are alternative views as well
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19 **Participants - Number**
- In fact, more studies revealed 5 not a good number and we need to employ more participants
  - Unfortunately, studies could not converge to a single number
- What we get from the literature is a range instead, between 5 and 25, both inclusive

20 **Participants - Number**
- For any pilot study (or small-scale studies), 5 is probably a good number
- However, the conclusions drawn from the data may be treated as indicative and more studies may be required

21 **Participants - Number**
- To draw *reliable* conclusions from empirical data, we should use between 12 and 25

participants (more the better)

22 **Note**

- Numbers are suggestive only - you can employ more participants
- However, it is less likely to get different conclusion(s) with more numbers than the ones already got with numbers suggested

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Task Design

24 **Tasks**

- In order to observe, we ask participants to perform *tasks*
  - E.g., typing task in our example
- It is very important to decide appropriate tasks - otherwise, observed data may not be useful
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25 **Tasks**

- An interface may support large number of tasks
- It is not necessary to ask participants to perform all these tasks - that may be impractical

26 **Tasks**

- Instead, participants should be given a set of representative tasks
  - Ex – "representative text" for typing task

27 **Assigning Tasks**

- Reconsider empirical study for RQ3
  - 2 test conditions (two interfaces)
  - We should design at least two tasks and ask participants to perform these tasks
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28 **Assigning Tasks**

- All the five users performed tasks for both the interfaces
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29 **Assigning Tasks**

- When each participant in a study performs tasks corresponding to all test conditions, we call the study design as "within-subject" or "repeated-measure"
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30 **Assigning Tasks**

- We cannot be so always, though
- Let's assume number of test conditions (interfaces) = 10
- Participants might object (or even refuse) to perform tasks (text entry) for so many interfaces!

31 **Assigning Tasks**

- TWO ways to resolve

32 **Assigning Tasks**

- We may introduce *breaks* in between task sessions for each participant

33 **Assigning Tasks**

- Ex - a participant performs tasks over a period of 2 days
  - First day, 5 interfaces (a manageable number)
  - Rest performed on the next day (or may be a different session in the same day after a gap of few hours)

34 **Assigning Tasks**

- Otherwise, we may ask each participant to perform text entry tasks for *only a subset* of the interfaces

35 **Assigning Tasks**

- Suppose we have 20 participants
- We divide them into 4 groups (5 in each)
- We ask each group to perform tasks *only* for 5 interfaces
- Each participant in a group performs the tasks for *all* 5 interfaces

36 **Assigning Tasks**

- When we distribute the tasks to participants in the way explained above, we call the study as "between-subject"

37 **Assigning Tasks**

- Within-subject designs are easier to manage - however, practice effect may happen
- It is not easy to decide on the right design (within-subject or between-subject)

38 **Assigning Tasks**

- Decision depends on availability of participants (typically, between-subject design requires more participants)
  - Even if participants are available, they might not be able to participate for long (within-subject designs requires longer involvement of the participants)
- Also depends on availability of other resources (study assistants, computers, laboratory space and so on)
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39 **Assigning Tasks**

- It is necessary to carefully consider these issues and balance the trade-offs

40 **Assigning Tasks**

- To avoid practice effect in within-subject design, we counterbalance sequence in which tasks are given to the participants
  - Task sequence for each participant is different from the other participants

41 **Assigning Tasks**

- One way - *randomize* the sequence for each participant

42 **Assigning Tasks**

- There is another, more systematic, method known as the Latin Square Method
- In this method, we organize sequence of tasks given to the participants in the form

of a square matrix, with the condition that each task occurs only once in each row and column

43 **Assigning Tasks**

- Ex – let's assume there are 4 participants (numbered P#1 to P#4)
- Each of them performs the text entry tasks for 4 interfaces
- Thus, each performs 4 tasks numbered R1 to R4

44 **Assigning Tasks**

45 **Book**

- Bhattacharya, S. (2019). Human-Computer Interaction: User-Centric Computing for Design, McGraw-Hill India
  - Print Edition: ISBN-13: 978-93-5316-804-9; ISBN-10: 93-5316-804-X
  - E-book Edition: ISBN-13: 978-93-5316-805-6; ISBN-10: 93-5316-805-8
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Chapter 7